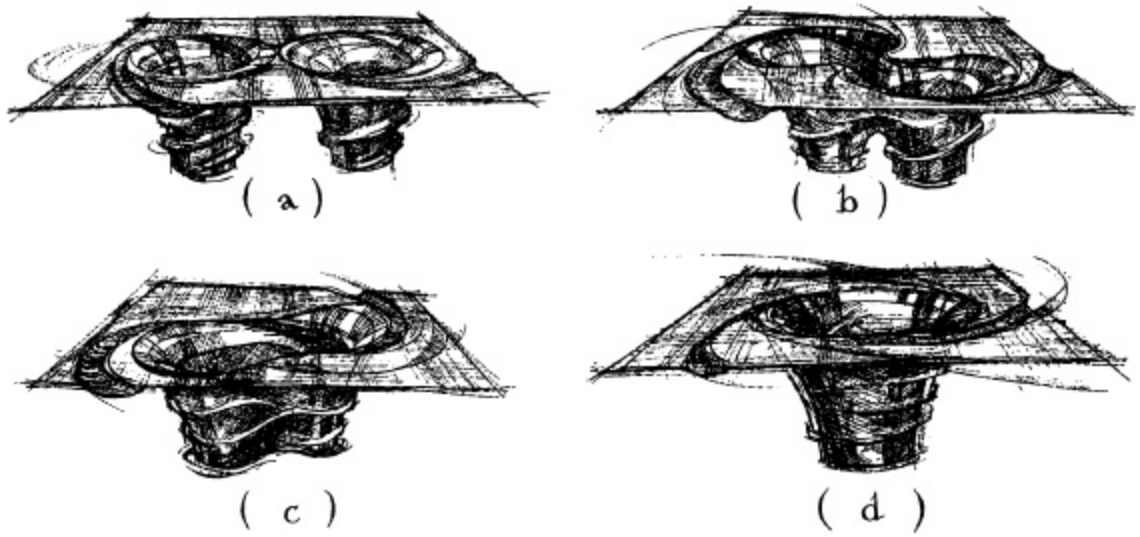
10.2a,b). Ultimately, when each hole is moving at nearly the speed of light, their horizons touch and merge. Where once there were two holes, now there is one—a rapidly spinning, dumbbell-shaped hole (Figure 10.2c). As the horizon spins, its dumbbell shape radiates ripples of curvature, and those ripples push back on the hole, gradually reducing its dumbbell protrusions until they are gone (Figure 10.2d). The spinning hole's horizon is left perfectly smooth and circular in equatorial cross section, with precisely the shape described by Kerr's solution to the Einstein field equation (Chapter 7).

By examining the final, smooth black hole, one cannot in any way discover its past history. One cannot discern whether it was created by the coalescence of two smaller holes, or by the direct implosion of a star made of matter, or by the direct implosion of a star made of antimatter. The black hole has no "hair" from which to decipher its history (Chapter 7).




**10.2 Embedding diagrams depicting the curvature of space around a binary system made of two black holes. The diagrams have been embellished by the artist to give a sense of motion. Each successive diagram is at a later moment of time, when the two holes have spiraled closer together. In diagrams (a) and (b), the holes' horizons are the circles at the bottoms of the pits. The horizons merge just before diagram (c), to form a single, dumbbell-shaped horizon. The rotating dumbbell emits gravitational waves, which carry away its deformation, leaving behind a smooth, spinning, Kerr black hole in diagram (d).** [Courtesy LIGO Project, California Institute of Technology.]

However, the history is not entirely lost. A record has been kept: It has been encoded in the ripples of spacetime curvature that the coalescing holes emitted. Those curvature ripples are much like the sound waves from a symphony. Just as the symphony is encoded in the sound waves'